INCH-POUND

A-A-52427
February 12, 1993
SUPERSEDES
MS51983E
30 August 1982

COMMERCIAL ITEM DESCRIPTION

NUT, PLAIN, SINGLE BALL SEAT, HEXAGON WHEEL MOUNTING - 3/4 AND 1-1/8 INCH

The General Services Administration has authorized the use of this commercial item description as a replacement for MS51983E which is canceled.

1.0 Abstract. This commercial item description (CID) covers a hexagon wheel mounting plain, single ball seat nut which is used to mount and secure wheels to the vehicle's axle and is referred to herein as the nut.

2.0 Salient characteristics.

2.1 Materials. The nut shall be made of carbon steel in accordance with SAE J995-88 Grade 5, and case hardened to a surface hardness of Rockwell 15N - 85.5 and to a depth of 0.005 to 0.012 inches or alloy steel that is through hardened to produce a core hardness reading of Brinell 223-277 and to a surface hardness of Rockwell 15N-69.5-84. The surface shall be zinc plated per ASTM B633, Type II, Class SC3 (FE/Zn13), except threads which are not fully plated. The use of recovered material made in compliance with regulatory requirements is acceptable providing that all requirements of this CID are met (see 5.5).

2.2 Design and construction. Unless otherwise specified in figure 1, the nut shall be designed and constructed in accordance with SAE J694-88 and shall be capable of sustaining the following proof load requirement without failure or permanent set. The proof loads shall be verified in accordance with SAE J995-79.

a. 3/4 size nut shall have a minimum proof load test of 37,000 pounds (lbs) of tensile force.

b. 1-1/8 size nut shall have a minimum proof load test of 87,000 lbs tensile force.

c. The nut shall be removeable by hand after load is released.

Beneficial comments, recommendations, additions, deletions clarifications, etc. and any other data which may improve this document should be sent by letter to: U.S. Army Tank-Automotive Command, ATTN: AMSTA-GDS, Warren, MI 48397-5000.

AMSC N/A FSC 5310

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NOTE: It may be necessary to use a manual wrench or other means to start nut in motion after load has been released to remove nut. Such wrenching is permissible providing that it is restricted to one half turn.

2.3 Markings. Markings shall be as specified in figure 1.

2.4 Workmanship. The nut shall be free from irregularities such as burrs, cracks, sharp edges, or any other defects affecting serviceability or appearance.

3.0 Quality assurance provisions.

3.1 Responsibility for inspection. The contractor is responsible for the performance of all inspection requirements (examinations and tests).

3.2 Contractor certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this commercial item description and that the product conforms to the producer's own drawings, specifications, standards, and quality assurance practices. Items with known defects shall not be submitted for Government acceptance. The Government reserves the right to require proof of such conformance prior to the first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

4.0 Preservation, packaging, packing, labeling, and marking. Preservation, packaging, packing, labeling, and marking shall be as specified in the contract or order (see 5.2).

5.0 Notes.

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

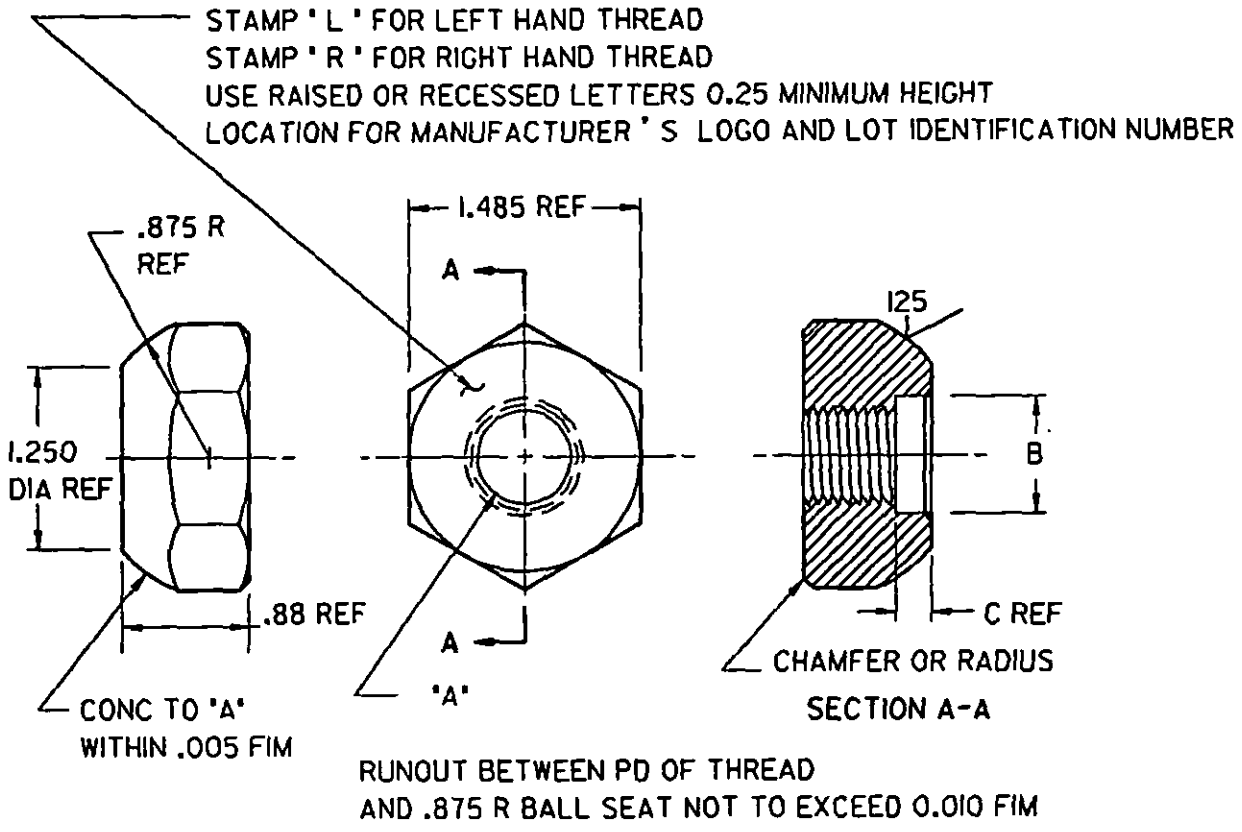
5.1 Addresses for obtaining copies of referenced documents.

5.1.1 Non-Government publications. ASTM B633-85 "Standardization Specification for Electrodeposited Coatings of Zinc on Iron and Steel"; can be obtained from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187. SAE J995-79 "Mechanical and Material Requirements for Steel Nuts, Standards" and SAE J694-88 "Disc Wheel/Hub or Drum Interface Dimensions - Commercial Vehicles" are available from the the Society of Automotive Engineers, Inc. (SAE), 400 Commonwealth Drive, Warrendale, PA 15096.

5.2 Ordering data. Acquisition documents must specify the following:

a. Title and date of this CID.
b. Issue of the DODISS to be cited in the solicitation and and the specific issue of individual documents referenced (see 5.1).
c. Part or identification number [PIN (see figure 1 and 5.3)].
d. Selection of applicable level and packaging requirements (see 4.0).

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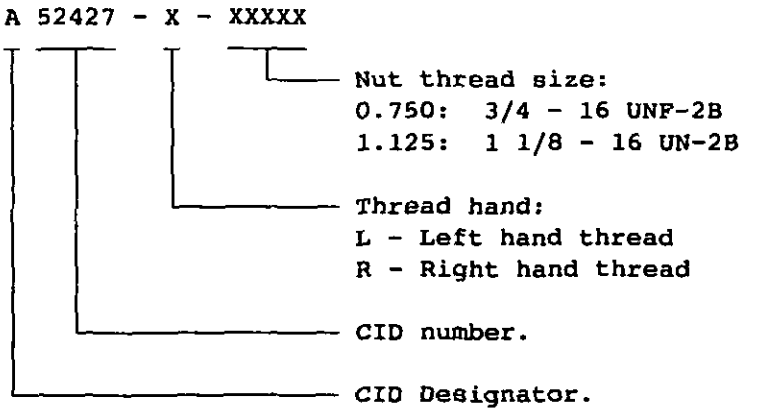
NOTE: Referenced dimensions shall be in accordance with SAE J694.

PIN no.	Former alloy steel part MS no.	Former steel part no.		Thread		Counter-bore	
		MS part no.	Drawing part no.	Hand	Designation "A"	"B"	"C"
A52427-L-0.750	-1	MS51983-5	537802	Left (L)	3/4-16-UNF-2B	0.78 ±.012	.280 ±.030
A52427-R-0.750	-2	MS51983-6	537801	Right (R)			
A52427-L-1.125	-3	MS51983-7	537806	Left (L)	1 1/8-16-UN-2B	1.16 ±.012	.170 ±.020
A52427-R-1.125	-4	MS51983-8	537805	Right (R)			

FIGURE 1. Wheel mounting nut design, configuration, and part number.

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5.3 CID based part identification number (PIN). The following part identification numbering procedure is for Government purposes and does not constitute a requirement for the contractor (see figure 1).



5.4 Cross-reference. Nuts conforming to this CID are interchangeable/ substitutable with nuts conforming to MS51983E, dated 30 August 1982.

5.5 Regulatory requirements. The offeror/contractor is encouraged to use recovered material in conformance with Public Law 94-580 to the maximum extent practicable.

5.6 Metric products. Wheel mounting nuts made to metric dimensions are subjected to the following requirements:

a. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within specified tolerances using conversion tables contained in the latest revision of ASTM E380, and all other requirements of this Commercial Item Description are met.
b. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable.
c. The contracting officer has the option of accepting or rejecting the product.

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